Specular Bit Architecture (SBA) for Bio-Photonic Neuromorphic Chips and DNA Storage: A Mathematical and Scientific Framework

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Abstract—This paper presents a substrate-agnostic formal framework for the Specular Bit Architecture (SBA), an informational layer for bio-photonic neuromorphic processors and DNA storage systems. SBA encodes signed information with specular carriers (without an explicit sign bit), uses Fibonacci positional weights for exact local rewrites, and defines deterministic sanitization and reconstruction with bounded propagation. The framework provides: (1) a signed-digit model with Zeckendorf-style uniqueness per channel; (2) exact rewrite rules derived from Fibonacci identities; (3) synchronous read-compute-write semantics with deterministic local arbitration (Protocol C); (4) exhaustive width-5 local enumeration to certify transition classes and extract bounded-propagation constants; and (5) an informational mass–energy bookkeeping model with an optional discrete–continuous geometric extension for propagation-resource estimation. Formal informational results are separated from speculative physical interpretations: parameters such as m_0 and c are modeling constants and require independent calibration for physical use. Under the stated synchronous and local-arbitration assumptions, the resulting primitives provide strict locality, deterministic normalization, and bounded error propagation across heterogeneous substrates, and admit a continuum-limit covariant conservation law (Proposition 10.1).

Index Terms—bio-photonic neuromorphic chips, bounded propagation, DNA data storage, Fibonacci positional weights, informational mass–energy, sanitization policy, signed-digit encoding, Specular Bit Architecture, Zeckendorf representation.

1 Introduction

Emerging high-density and low-latency substrates — including photonic neuromorphic arrays, hybrid bio-electronic microcircuits, and DNA archival media — motivate an informational layer that maximizes locality, tolerates transient ambiguous states, and provides deterministic, reproducible semantics for local correction. The Specular Bit Architecture (SBA) is proposed as such a layer: an encoding and execution model in which information is generated simultaneously in two opposite directions and represented as a unified vector state. SBA departs from conventional scalar bit encodings by replacing an explicit sign bit with *specular carriers*, thereby enabling signed-digit representations with local, exact rewrite rules and deterministic sanitization.

The principal contributions of this work are:

1) A formal definition of SBA primitives and carrier semantics suitable for multiple physical substrates, including a mapping that naturally fits DNA base encodings and photonic carriers.
2) A family of Fibonacci positional weights $W_i = F_{i+o}$ and an algebraic identity $2W_i = W_{i+1} + W_{i-2}$ that underpin exact local rewrites and guarantee weighted-sum preservation.
3) A synchronous round execution model (Read, Compute, Write, Sanitization, Scheduling & Arbitration, Commit) with a deterministic, local arbitration protocol (Protocol C) that ensures at most one committed reconstruction per target per round.
4) A discrete variational formulation (Section 7.1) in which SBA updates are local minimizers of an information action under Protocol C weights.
5) Proofs of canonical uniqueness, lexicographic termination, and bounded propagation under the synchronous model and Protocol C.
6) An exhaustive width-5 local enumeration framework that certifies transition classes and supports extraction of bounded-propagation constants from a finite transition table.
7) An informational mass–energy bookkeeping model that provides register-level cost metrics, together with an optional discrete–continuous geometric extension for propagation-resource estimation.
8) A continuum-limit conservation statement linking SBA round dynamics to emergent geometric consistency, formalized by Proposition 10.1 in Section 10.9.

SBA is explicitly presented as a formal informational framework: all theorems, lemmas, and operational protocols remain independent of any particular physical realization. Where physical interpretations are discussed (e.g., mapping carriers to nucleobases or interpreting the radiative energy register), they are labeled as engineering hypotheses and do not affect the formal correctness of the informational model. For visibility, the discrete CPT invariance result is stated as Theorem 7.2 in Section 7.1.

1.1 Claim classes and evidentiary status

To avoid category errors in interpretation, claims in this manuscript are partitioned into three classes.

- **Formal claims (proved):** algebraic identities, canonical uniqueness, exact local preservation, lexicographic termination, and bounded propagation results proved under the explicitly stated synchronous semantics and Protocol C assumptions.

- **Modeling claims (derived):** informational mass–energy bookkeeping and discrete–continuous bridge equations that are mathematically consistent with the formal layer but introduce calibration constants (e.g., $m_0, \mu, c_I, G_I, \alpha$).
- **Physical claims (hypothesis-level):** substrate-specific and deep-space predictions that require independent empirical calibration and uncertainty-quantified experiments before any physical interpretation is accepted.

Accordingly, agreement with one claim class does not, by itself, validate stronger claims in another class.

1.2 Motivation and design goals

The SBA design is guided by four engineering desiderata:

- **Locality:** all normalization and reconstruction operations must be local (bounded radius) to enable scalable hardware implementations and to limit error propagation.
- **Determinism:** given the same local snapshot, reconstruction decisions must be deterministic to avoid non-deterministic divergence and race conditions.
- **Exactness:** local rewrites must preserve the signed weighted sum exactly, enabling lossless normalization and canonical representations.
- **Substrate agnosticism:** the informational primitives must be implementable on diverse physical carriers (digital registers, DNA oligos, photonic waveguides) with a clear mapping to substrate-specific constraints.

1.3 Paper organization

Section 2 introduces notation and carrier layout. Section 3 formalizes Fibonacci positional weights and the algebraic identity used for exact local rewrites. Section 4 proves uniqueness properties for canonical representations. Section 5 defines synchronous round semantics, interior and boundary rewrites, and sanitization/reconstruction. Section 7 presents Protocol C and its guarantees. The variational minimum-action formulation based on a discrete information action is introduced in Section 7.1. Section 8 introduces the lexicographic measure and proves bounded propagation. Section 9 defines informational mass–energy bookkeeping, and Section 10 presents an optional discrete–continuous geometric extension. Appendices collect proofs, exhaustive-enumeration artifacts, and implementation notes.

2 Notation and Physical Carrier Layout

We collect here the principal notation and the carrier semantics that implementations must respect. The presentation is intentionally substrate-agnostic: the same logical primitives apply to digital registers, DNA oligo blocks, or photonic carriers provided the physical mapping satisfies the carrier constraints below.

Definition 2.1 (Physical carrier per position)**.** *For each sequence position i we define two logical SBA channels A and B. Each channel is represented by a two-state physical carrier pair:*

$$\textit{Channel } X \in \{A, B\}: \quad (p_i^{(X)}, n_i^{(X)}),$$

where p denotes the positive pole and n the negative pole. The logical digit decoded from a carrier is the differential

$$\delta(p, n) := p - n \in \{-1, 0, +1\},$$

with valid physical micro-patterns corresponding to $00, 01, 10$. The pattern 11 is reserved as a transient ambiguous microstate and must be sanitized deterministically.

Remark 2.2 (Substrate mapping for DNA as a mirror hypothesis)**.** *In DNA storage mappings, the four physical bits required per positional index i can be assigned to the four nucleobases. One admissible assignment uses Adenine/Thymine for Channel A and Cytosine/Guanine for Channel B. Under this assignment, DNA is modeled not as a purely scalar biological string but as a dual-channel carrier implementation of SBA-style vectorial information.*

A speculative geometric interpretation is to treat complementary base pairing and strand anti-parallelism as a local mirror architecture around a central axis ("central zero" in the model vocabulary): e.g., complementary microstates can be encoded as specular pairs $01 \leftrightarrow 10$. In this interpretation, matter/antimatter language is used only as an analogy for opposite signed informational carriers, not as a biological claim about literal particle content in DNA.

Accordingly, this remark introduces an engineering/cosmological mapping hypothesis for substrate design intuition; it does not alter the formal correctness of the SBA algebra, rewrite rules, or proofs.

2.0.0.1 Glossary of symbols:

- $W_i^{(X)}$: positional weight at index i for channel $X \in \{A, B\}$.
- $a_i^{(X)}, b_i^{(X)}$: canonical digits in channel X.
- $d_i^{(X)}$: intermediate digit sum $a_i^{(X)} + b_i^{(X)}$.
- S: sanitization operator mapping transient $11 \mapsto 00$.
- σ: specular swap operator on carriers.
- $R_i^{(X)}$: deterministic reconstruction function at index i.
- $K_i^{(X)}$: sanitization flag at index i.
- E_{rad}: radiative energy register (modeling analog propagation resources).
- m_0: model unit mass in the discrete bookkeeping mass $M(V) = m_0 \sum_i |v_i| W_i$ (Section 9).
- m_{SBA}: dimensionless occupancy index $\sum_i |a_i| W_i$ used in the dimensionalized SBA formulation (Equation (6)).
- m_{phys}: physical mass proxy defined by $m_{\text{phys}} = \mu m_{\text{SBA}}$ (Equation (6)).
- μ: calibration constant (kg) mapping m_{SBA} to m_{phys}.
- $R(d), m(d), p(d)$: lexicographic measure components (rightmost overflow index, multiplicity, pair).
- o_A, o_B: integer offsets for channel weights, with $o_A, o_B \geq 2$ in the default admissible setting.

3 Fibonacci Positional Weights and Algebraic Identity

Let $(F_k)_{k\in\mathbb{Z}}$ denote the Fibonacci sequence with $F_0 = 0, F_1 = 1$, extended to negative indices by

$$F_{-n} = (-1)^{n+1} F_n, \qquad n \ge 1.$$

For each channel X fix an integer offset $o_X \ge 2$ (admissible offset regime) and define positional weights

$$W_i^{(X)} := F_{i+o_X}, \qquad i \in \mathbb{Z}. \tag{1}$$

When the channel is clear from context we write W_i. This convention makes boundary/guard-index expressions (e.g., $i-2$) formally well-defined while guard-cell digit values remain fixed by the execution semantics.

For $i \ge 2$ these weights satisfy the identity

$$2W_i = W_{i+1} + W_{i-2}, \tag{2}$$

which is the algebraic basis for the exact local rewrite rules used to normalize intermediate digit sums.

Remark 3.1 (Biological interpretation of weights)**.** *When applied to macromolecular substrates such as DNA, these Fibonacci weights define a hierarchical positional scale: index importance grows with position according to Fibonacci scaling, yielding local correction rules with bounded neighborhood dependence.*

4 Canonical Form and Uniqueness

Definition 4.1 (Canonical sequence)**.** *A digit sequence $x = (x_i)$ with $x_i \in \{-1, 0, 1\}$ is* canonical *if for every i it holds $x_i \cdot x_{i+1} = 0$ (no adjacent nonzero digits).*

Lemma 4.2 (Zeckendorf uniqueness for positive support)**.** *For weights $W_i = F_{i+o}$ with $o \ge 2$, and digits $x_i \in \{0, 1\}$ with the non-adjacency constraint, every nonnegative integer admits a unique representation $\sum_i x_i W_i$ (shifted Zeckendorf form).*

Lemma 4.3 (Signed canonical uniqueness via specular carriers)**.** *Let c be a canonical carrier sequence. Define the specular swap operator σ that exchanges p_i and n_i for each carrier. Then $\delta(\sigma(c_i)) = -\delta(c_i)$ and σ is involutive. If two canonical carrier sequences have the same weighted value $\sum_i \delta(c_i) W_i$, then they coincide pointwise; hence the signed canonical representation is unique.*

Proposition 4.4 (Uniqueness of dual independent streams)**.** *Let $W_i^{(A)} = F_{i+o_A}$ and $W_i^{(B)} = F_{i+o_B}$ with $o_A, o_B \ge 2$. Under the canonical non-adjacency constraint applied independently to each channel, the weighted representation is unique per channel. Consequently, the two-component logical state $\mathbf{V} = [V_A\ V_B]^T$ is unique.*

5 Intermediate Representation and Exact Local Rewrites

Given two canonical operands $A^{(X)}$ and $B^{(X)}$ for channel X, define the intermediate digit array

$$d_i^{(X)} := a_i^{(X)} + b_i^{(X)} \in \{-2, -1, 0, 1, 2\}. \tag{3}$$

5.1 Execution model: synchronous round semantics

Normalization is defined independently for each channel and proceeds in synchronous rounds with the following ordered phases:

1) **Read:** read a latched snapshot of current digits $d^{(X)}$.
2) **Compute:** compute candidate interior rewrites for indices with $|d_i^{(X)}| = 2$ using (2).
3) **Write:** commit computed rewrites atomically (single linearization point τ_W).
4) **Sanitization:** apply S to physical carriers; set flags $K_i^{(X)}$ for $11 \mapsto 00$ events.
5) **Scheduling & Arbitration:** for each flagged index schedule compensation candidates; apply Protocol C deterministic arbitration to select at most one committed injection per target.
6) **Commit:** commit selected compensating injections and advance to the next round.

Let n denote the register length (active indices $0, \dots, n-1$). Virtual guard cells $d_{-2}^{(X)} = d_{-1}^{(X)} = d_n^{(X)} = d_{n+1}^{(X)} = 0$ enforce boundary semantics for each channel.

Definition 5.1 (Round consistency)**.** *A round is* consistent *if all rewrite and reconstruction decisions are computed from the same pre-round snapshot, all writes are committed atomically at τ_W, and sanitization/arbitration outcomes are applied only after that commit.*

5.2 Interior rewrite rules (indices $i \ge 2$)

Suppressing channel superscripts for brevity ($d_i = d_i^{(X)}$, $W_i = W_i^{(X)}$): if $d_i = +2$, apply the exact local update

$$d_i \leftarrow d_i - 2, \quad d_{i+1} \leftarrow d_{i+1} + 1, \quad d_{i-2} \leftarrow d_{i-2} + 1. \tag{4}$$

If $d_i = -2$, apply the symmetric update

$$d_i \leftarrow d_i + 2, \quad d_{i+1} \leftarrow d_{i+1} - 1, \quad d_{i-2} \leftarrow d_{i-2} - 1. \tag{5}$$

Both updates are local, exact, and preserve the signed weighted sum $\sum_i d_i W_i$ by (2).

5.3 Boundary rules

For $i = 0$: if $d_0 = +2$, set $d_0 \leftarrow d_0 - 2$ and $d_1 \leftarrow d_1 + 1$ (symmetric rule for -2). For $i = 1$: if $d_1 = +2$, set $d_1 \leftarrow d_1 - 2$, $d_2 \leftarrow d_2 + 1$, and $d_0 \leftarrow d_0 + 1$ (symmetric rule for -2). These boundary updates preserve the weighted sum and cannot create an overflow strictly to the right of the pre-round rightmost overflow index $R(d)$.

5.4 Informational Mass, Dimensional Coherence, and the $E = mc^2$ SBA Formulation

To connect the informational model to measurable quantities, we introduce a dimensional scaling for SBA states while preserving the formal substrate-agnostic interpretation adopted throughout this manuscript. Let $a_i \in \{-1, 0, +1\}$ denote the signed state of register i, where $+1$ maps to the canonical carrier (`01`), -1 maps to the specular carrier (`10`), and 0 denotes inactivity.

Because Fibonacci weights $W_i = F_{i+o}$ are dimensionless, dimensional coherence is obtained by introducing a calibration constant μ with SI unit kg. This μ is separate from the discrete bookkeeping constant m_0 of Section 9. Define the dimensionless informational mass index

$$m_{\mathrm{SBA}} := \sum_i |a_i| W_i,$$

and the corresponding physical mass proxy

$$m_{\mathrm{phys}} = \mu\, m_{\mathrm{SBA}} = \mu \sum_i |a_i| W_i. \tag{6}$$

A model-level total energy functional is then defined as

$$\begin{aligned} E_{\mathrm{tot}} &= m_{\mathrm{phys}} c^2 + U_{\mathrm{bind}}(\{a_i\}) \\ &= \left(\mu \sum_i |a_i| W_i\right) c^2 + U_{\mathrm{bind}}(\{a_i\}). \end{aligned} \tag{7}$$

Here, U_{bind} captures non-additive interaction terms of multi-register configurations.

In this interpretation:

1) **Role of c:** c is the speed of light used in the mass–energy conversion factor; any additional interpretation as a system refresh-rate parameter is a modeling hypothesis, not a derived physical identity.
2) **Role of $|a_i|$:** the mass proxy depends on occupancy magnitude, so $+1$ and -1 contribute equally to m_{phys}, consistent with equal positive rest-mass contributions in the model.
3) **Role of U_{bind}:** this term represents configuration-dependent corrections (analogous in spirit to binding-energy terms) and requires independent calibration for any specific substrate.

Spin box — single-register activation scale (hypothetical calibration). Assume one active register with $|a_i| = 1$, negligible U_{bind}, and $\mu = 10^{-30}\,\mathrm{kg}$ (illustrative). The activation contribution at index i is

$$E_i \approx \mu W_i c^2.$$

For example, if $W_i = F_9 = 34$, then

$$E_i \approx 10^{-30} \cdot 34 \cdot (3 \times 10^8)^2 \approx 3.06 \times 10^{-12}\,\mathrm{J}.$$

So a single activation at that weight is on the order of picojoules.

5.4.0.1 Annihilation rule (model-level cancellation): If two opposite signed contributions at the same effective degree of freedom cancel (net state driven to 0), the occupancy term $\sum_i |a_i| W_i$ decreases. The corresponding modeled released energy is

$$\Delta E = \mu \left(\sum_i |a_i| W_i\right)_{\text{before}} c^2 - \mu \left(\sum_i |a_i| W_i\right)_{\text{after}} c^2, \tag{8}$$

with any emitted-radiation interpretation treated as substrate-specific and external to the formal SBA proofs.

Definition 5.2 (Local composition and transient commit semantics)**.** *For each register i in one synchronous round, let N_i^+ and N_i^- denote the numbers of pre-commit positive and negative unit contributions targeting i after Scheduling & Arbitration and before Commit. Define:*

$$a_i^{\mathrm{pre}} := N_i^+ - N_i^-, \qquad m_i^{\mathrm{pre}} := (N_i^+ + N_i^-) W_i,$$
$$a_i^{\mathrm{post}} := \mathrm{sat}_{\{-1,0,1\}}(a_i^{\mathrm{pre}}), \qquad m_i^{\mathrm{post}} := |a_i^{\mathrm{post}}| W_i,$$

where $\mathrm{sat}_{\{-1,0,1\}}$ is clipping to $\{-1,0,1\}$. Under Protocol C (Lemma 7.5), $(N_i^+, N_i^-) \in \{(0,0),(1,0),(0,1)\}$ at commit time, so $a_i^{\mathrm{pre}} = a_i^{\mathrm{post}} \in \{-1,0,1\}$.

Operationally: if opposite carriers (`01` and `10`) are simultaneously present only as transient pre-commit candidates for the same target, they are counted separately in m_i^{pre}; after Commit the register is represented by the single net state a_i^{post}, so perfect local cancellation gives zero post-commit contribution.

For manuscript-level mass–energy accounting, m_{SBA} and m_{phys} use post-commit states:

$$m_{\mathrm{SBA}} = \sum_i |a_i^{\mathrm{post}}| W_i, \qquad m_{\mathrm{phys}} = \mu \sum_i |a_i^{\mathrm{post}}| W_i.$$

5.5 Units and Scale

The dimensionalized SBA mapping uses SI units as follows: m_{SBA} is dimensionless, μ has unit kg, m_{phys} has unit kg, and E_{tot}, ΔE have unit J. Therefore, μc^2 has unit J per unit of m_{SBA}.

A practical calibration workflow is:

1) Measure an effective energy increment ΔE_{meas} for a controlled operation whose modeled occupancy change is Δm_{SBA}.
2) Estimate μ by

$$\mu = \frac{\Delta E_{\mathrm{meas}}}{\Delta m_{\mathrm{SBA}}\, c^2},$$

then validate the estimate on independent configurations.

Numerical examples. If $\Delta m_{\mathrm{SBA}} = 1$ and $\mu = 10^{-30}\,\mathrm{kg}$, then

$$\Delta E \approx \mu c^2 \approx 9 \times 10^{-14}\,\mathrm{J}.$$

If $\Delta m_{\mathrm{SBA}} = 34$ (e.g., occupancy over Fibonacci-weighted sites summing to 34) with the same μ, then

$$\Delta E \approx 34\,\mu c^2 \approx 3.06 \times 10^{-12}\,\mathrm{J}.$$

These examples are scale illustrations only; substrate-specific μ must be determined experimentally.

6 Sanitization and Deterministic Local Reconstruction

The physical pattern 11 is transient and must be sanitized. Sanitization is modeled by operator S that maps $11 \mapsto 00$ and sets a reconstruction flag $K_i^{(X)}$. A deterministic reconstruction function $R_i^{(X)}$ examines a local neighborhood $d_{i-r}, \ldots, d_{i+r}$ ($r \leq 2$) and proposes a candidate injection $\tilde{v}_i \in \{-1,0,1\}$.

Timing convention: $R_i^{(X)}$ reads the post-sanitization snapshot (state after Write and Sanitization, before Scheduling & Arbitration); all reconstructors compute from the same latched snapshot for the round.

7 Protocol C: Deterministic Local Arbitration

Protocol C deterministically resolves competing candidate injections targeting the same index in a synchronous round.

7.0.0.1 Assumptions:

1) Locality and snapshot consistency: each candidate $v_{i\to j}$ originates from a source i with $|i-j| \le r$; all candidates are computed from the same latched snapshot.
2) Deterministic energy score: each candidate is assigned a locally computable score $E_{i\to j}$ depending only on the local neighborhood.
3) Deterministic tie-breaking: ties on $E_{i\to j}$ are broken deterministically (e.g., smaller source index).
4) Atomic target lock: committing a transfer to target j requires acquiring an atomic lock; at most one commit to j linearizes at τ_C per round.
5) Bounded candidate values: each candidate $v_{i\to j} \in \{-1, 0, 1\}$.

7.0.0.2 Per-target synchronization sequence: For each target j in a round:

1) **SCORE:** reconstructors compute $E_{i\to j}$ and proposals from snapshot s_t.
2) **SELECT:** arbiter j deterministically orders proposals by $(E_{i\to j}, \text{tie-break key})$ and emits winner id w_j.
3) **COMMIT:** only w_j may attempt lock acquisition; if successful, w_j commits at τ_C and releases the lock; non-winners are invalidated.

7.0.0.3 Admissible energy score template: A concrete admissible score is

$$E_{i\to j} := \alpha\, \Delta M^{\text{local}}_{i\to j} + \beta\, |i-j| + \gamma\, \mathbf{1}_{\{K_i=0\}}, \qquad \alpha, \beta, \gamma > 0,$$

where

$$\Delta M^{\text{local}}_{i\to j} := m_0 \sum_{k\in\mathcal{N}_r(j)} \Big(|d_k^{\text{after}(i\to j)}| - |d_k^{\text{before}}|\Big) W_k,$$

$\mathcal{N}_r(j) = \{k : |k-j| \le r\}$, and $\mathbf{1}_{\{K_i=0\}}$ penalizes candidates not sourced from a flagged sanitization site. This score is locally computable, deterministic, monotone in local mass change, totally orderable with deterministic tie-break, and $O(1)$ to evaluate for fixed r.

7.1 Discrete information action and minimum-action interpretation

To express SBA dynamics in variational form, define the round-state $s_t = (d_t, K_t, E_{\text{rad},t})$, where d_t is the signed-digit array, K_t is the sanitization-flag field, and $E_{\text{rad},t}$ is the radiative bookkeeping register.

Let u_t denote the set of legal round updates (interior/boundary Fibonacci rewrites and Protocol C commits) and let $s_{t+1} = F(s_t, u_t)$ be the synchronous round map. Over horizon $t = 0, \dots, T-1$, define the information action

$$\mathcal{A}[s_{0:T}, u_{0:T-1}] := \sum_{t=0}^{T-1} \mathcal{L}(s_t, u_t, s_{t+1}), \tag{9}$$

with per-round discrete Lagrangian

$$\mathcal{L} = \alpha\, \Delta\Phi_t + \beta\, \Delta H_t + \gamma\, \Delta Q_t + \lambda\, \mathcal{C}_t, \tag{10}$$

where $\alpha, \beta, \gamma > 0$, $\lambda \gg 1$, and

$$\Delta\Phi_t := \Phi(d_{t+1}) - \Phi(d_t), \tag{11}$$

$$\Phi(d) := \sum_i |d_i| W_i + \rho \sum_i \mathbf{1}_{\{|d_i|=2\}}, \quad \rho > 0, \tag{12}$$

$$\Delta H_t := H_{t+1} - H_t, \tag{13}$$

$$H_t := -\sum_{\sigma\in\{-1,0,1,\perp\}} p_t(\sigma) \log p_t(\sigma), \tag{14}$$

$$\Delta Q_t := Q_{t+1} - Q_t, \qquad \Delta Q_t \ge k_B T \ln 2\, N_{\text{erase},t}. \tag{15}$$

Here $\perp$ denotes the transient ambiguous microstate (physical pattern 11); $N_{\text{erase},t}$ counts irreversible sanitization/erasure events in round t. The constraint penalty $\mathcal{C}_t$ enforces legality (admissible digit alphabet, weighted-sum preservation under Fibonacci identities, single-winner Protocol C commit per target, and boundary/guard rules).

The minimum-action update at round t is

$$u_t^\star \in \arg\min_{u\in\mathcal{U}(s_t)} \mathcal{L}\big(s_t, u, F(s_t, u)\big), \tag{16}$$

where $\mathcal{U}(s_t)$ is the legal local-update set.

This formulation is consistent with the existing SBA mechanics:

1) Fibonacci rewrites are legal moves that preserve $\sum_i d_i W_i$ exactly and reduce overflow content through $\Delta\Phi_t$.
2) Protocol C arbitration with weights (α, β, γ) selects the locally minimal incremental cost among admissible candidates.
3) Sanitization reduces ambiguous-state occupancy (entropy-driving term) while accounting for irreversible erasure via the Landauer lower bound in (15).

Remark 7.1 (Physical interpretation as hypothesis)**.** *Equations* (9)–(16) *define a formal informational variational principle for SBA dynamics. Any stronger statement that physical substrates or "the universe" realize these updates as globally optimal heat-dissipation trajectories is an engineering/physics hypothesis requiring independent empirical calibration of (α, β, γ), entropy estimates, and effective erasure costs.*

Theorem 7.2 (Discrete CPT invariance of the SBA information action)**.** *Fix a finite centered index set $I = \{-N, \dots, N\}$ and synchronous rounds $t = 0, \dots, T$. Assume: (i) mirrored channels use the same Fibonacci weight law $W_i^{(A)} = W_{-i}^{(B)}$; (ii) Protocol C uses the same coefficients (α, β, γ) on both mirror channels; and (iii) the legal-update set is closed under round reversal. Define Θ_{CPT} on trajectories by*

$$\text{C}: \quad d_{i,t}^{(A)} \mapsto -d_{i,t}^{(B)}, \;\; d_{i,t}^{(B)} \mapsto -d_{i,t}^{(A)}, \tag{17}$$

$$\text{P}: \quad i \mapsto -i, \tag{18}$$

$$\text{T}: \quad t \mapsto T-t \text{ and } u_t \mapsto u^{\text{rev}}_{T-1-t}. \tag{19}$$

Then the transformed trajectory $\Theta_{\text{CPT}}(s_{0:T}, u_{0:T-1})$ is admissible and

$$\mathcal{A}\big[\Theta_{\text{CPT}}(s_{0:T}, u_{0:T-1})\big] = \mathcal{A}[s_{0:T}, u_{0:T-1}], \tag{20}$$

where $\mathcal{A}$ is defined in (9)*. Hence the minimum-action rule* (16) *is CPT-covariant.*

Proof. Under C, digits change sign and mirror channel while $|d_i|$, overflow indicators, and local candidate magnitudes are unchanged; therefore Φ and the local score magnitudes in Protocol C are preserved. Under P, index reflection together with $W_i^{(A)} = W_{-i}^{(B)}$ preserves weighted terms and neighborhood radii. Under T, reversed legal updates produce the same per-round increments with opposite ordering, so the sum in (9) is unchanged. Consequently each contribution in (10) is invariant: $\Delta\Phi_t, \Delta H_t, \Delta Q_t, \mathcal{C}_t$. Summing over t yields (20).

Because action values are equal on CPT-related admissible trajectories, argmin sets are mapped into argmin sets; therefore the minimum-action policy is CPT-covariant. □

Remark 7.3 (Interpretation)**.** *Theorem 7.2 is a formal invariance statement for the discrete SBA action under the mirror-channel* CPT *transform. Interpreting this invariance as a substrate-level physical conservation law requires independent experimental validation of the model assumptions.*

Lemma 7.4 (Total orderability and monotonicity of Protocol C scores)**.** *Fix a target j and let $\mathcal{C}_j$ be its finite candidate set in one round. For $\alpha, \beta, \gamma > 0$, define for each candidate $c = (i, v) \in \mathcal{C}_j$*

$$E(c) = \alpha\, \Delta M^{\mathrm{local}}(c) + \beta\, |i - j| + \gamma\, \mathbf{1}_{\{K_i = 0\}}.$$

If candidates are ranked by the deterministic key $(E(c), \tau(c))$, where τ is a deterministic tie-break key (e.g., source index, then candidate id), then:

1) ***Total orderability:*** *the induced order on $\mathcal{C}_j$ is total (every two candidates are comparable) and therefore has a unique minimum.*
2) ***Monotonicity in local mass term:*** *for fixed $|i - j|$, $\mathbf{1}_{\{K_i=0\}}$, and tie-break key, if $\Delta M^{\mathrm{local}}(c_1) < \Delta M^{\mathrm{local}}(c_2)$, then $E(c_1) < E(c_2)$.*

Proof. Because $\mathcal{C}_j$ is finite and τ is deterministic, (E, τ) maps each candidate to an element of a totally ordered set (lexicographic order on $\mathbb{R} \times T$, where T is totally ordered). Hence the induced relation is total and admits a unique minimum.

For monotonicity, hold $|i - j|$, $\mathbf{1}_{\{K_i=0\}}$, and τ fixed. Then

$$E(c_2) - E(c_1) = \alpha\big(\Delta M^{\mathrm{local}}(c_2) - \Delta M^{\mathrm{local}}(c_1)\big),$$

which is positive whenever $\Delta M^{\mathrm{local}}(c_2) > \Delta M^{\mathrm{local}}(c_1)$, since $\alpha > 0$. Therefore the score is strictly increasing in $\Delta M^{\mathrm{local}}$ under those fixed terms. □

Lemma 7.5 (Protocol C non-accumulation)**.** *Under Assumptions 1–5, Protocol C commits at most one transfer to any target j per round; hence the sum of committed transfers to j lies in $\{-1, 0, 1\}$.*

8 Lexicographic Measure and Bounded Propagation

Define

$$R(d) := \max\{i : |d_i| = 2\}, \qquad m(d) := |\{i : |d_i| = 2\}|\,. \tag{21}$$

The lexicographic measure $p(d) = (R(d), m(d))$ is well-founded on finite configurations and strictly decreases lexicographically under the synchronous round semantics combined with Protocol C.

Theorem 8.1 (Bounded propagation)**.** *Under synchronous read-then-write semantics, canonical inputs per channel, and Protocol C for reconstruction arbitration, any initial overflow event $|d_i| = 2$ is resolved within a neighborhood of radius $R_{\max}$ independent of the global length n. The number of synchronous rounds required to eliminate all ± 2 values is bounded by a constant K independent of n. A conservative explicit linear bound in terms of R_0 and $m(0)$ is given in Proposition 8.2.*

Proof. Fix a synchronous round t, and let $d^{(t)}$ denote the pre-round signed difference field. Let

$$\begin{aligned} p\big(d^{(t)}\big) &:= \big(R_t, m_t\big), \\ R_t &:= \max\{i : |d_i^{(t)}| = 2\}, \\ m_t &:= \#\{i : |d_i^{(t)}| = 2\}. \end{aligned}$$

Assume $m_t > 0$; otherwise no overflow exists and the claim is already true.

By the synchronous read-then-write semantics, all proposals in round t are computed from $d^{(t)}$, and their effects are committed simultaneously at the end of the round. Therefore, no newly written value in round t can trigger a second rewrite in the same round.

Let i be any index with $|d_i^{(t)}| = 2$. By (4)–(5), an interior rewrite at i:

1) removes the overflow at i, i.e., $|d_i| = 2 \to 0$ at the acted site;
2) changes only the local stencil $\{i-2, i, i+1\}$;
3) injects only unit-magnitude updates (each contribution is in $\{-1, 0, +1\}$).

Hence a single acted overflow cannot directly create $|d_k| = 2$ outside a bounded local radius.

Next consider sanitation/reconstruction. By Protocol C (Lemma 7.5), for each target index j, at most one transfer to j is committed in a given round; thus the net committed transfer into j is in $\{-1, 0, +1\}$. Consequently, concurrent proposals cannot accumulate in one round to create a jump of magnitude 2 at the same target.

Now compare $p\big(d^{(t+1)}\big)$ to $p\big(d^{(t)}\big)$:

- If an overflow at index R_t is acted in round t, then that site is cleared by the rewrite rule, and no same-round cascade can recreate a farther overflow to the right (synchronous linearization). Therefore $R_{t+1} < R_t$, or $R_{t+1} = R_t$ only if another pre-existing overflow at R_t remained, which still decreases m_t once one is removed.
- If $R_{t+1} = R_t$, then the number of ± 2 sites at this maximal radius decreases because each committed action clears one acted overflow and Protocol C blocks multiplicity-based re-creation at one target in the same round. Hence $m_{t+1} < m_t$.

So in all nonterminal rounds,

$$p\big(d^{(t+1)}\big) \;<_{\mathrm{lex}}\; p\big(d^{(t)}\big).$$

Because $p(d) \in \mathbb{N} \times \mathbb{N}$ with lexicographic order is well-founded on finite configurations, strict descent cannot continue indefinitely; therefore after finitely many rounds there is no index with $|d_i| = 2$.

Finally, locality gives bounded propagation: each committed update is confined to fixed-radius stencils (interior radius from (4)–(5) plus reconstruction radius r), so influence from any initial overflow cannot escape a finite neighborhood whose radius depends only on rule radii, not on global length n. Since each round strictly decreases p, the total number of rounds is bounded by a constant K determined by those local bounds and the local initial overflow multiplicity envelope, independent of n. This proves bounded neighborhood and bounded-round resolution. □

Proposition 8.2 (Conservative numerical round bound)**.** *Let $d^{(0)}$ be the initial difference field, with*

$$R_0 := R(d^{(0)}), \qquad m(0) := m(d^{(0)}).$$

Define the effective local radius

$$\rho := \max\{2, r\},$$

where 2 is the maximal interior-rewrite reach from (4)–(5) *and r is the reconstruction neighborhood radius. Under the synchronous model and Protocol C assumptions, a conservative bound is*

$$K \leq c_1 R_0 + c_2 m(0), \qquad c_1 := \rho + 1, \quad c_2 := \rho. \tag{22}$$

Hence

$$K \leq (\rho + 1)R_0 + \rho\, m(0).$$

Proof sketch. The lexicographic argument ensures strict descent of $p(d) = (R, m)$ each nonterminal round. To obtain explicit coefficients, upper-bound the latency of one lexicographic decrement by a radius-only constant.

Because updates are local and synchronous, a dependency can traverse at most one local neighborhood per round; with effective radius ρ, any required local coordination for clearing a right-edge overflow is completed within at most $\rho + 1$ rounds (ρ rounds for neighborhood interaction plus one commit round). Thus each unit decrease of R costs at most $\rho + 1$ rounds.

At fixed right edge, Protocol C non-accumulation prevents multi-source same-round re-creation at one target; clearing one remaining overflow multiplicity at that edge requires at most ρ rounds. Summing worst-case costs over at most R_0 right-edge decreases and $m(0)$ multiplicity removals yields (22). □

8.0.0.1 How to compute c_1, c_2 from rule radii.: Use the following recipe for a fixed rule set: (i) compute rewrite reach $\rho_{\mathrm{rw}} := \max\{|\Delta| : d_{i+\Delta}$ can be modified by one rewrite at $i\}$ (here $\rho_{\mathrm{rw}} = 2$); (ii) take reconstruction radius $\rho_{\mathrm{rec}} := r$; (iii) set $\rho = \max\{\rho_{\mathrm{rw}}, \rho_{\mathrm{rec}}\}$; (iv) use $c_1 = \rho + 1$, $c_2 = \rho$. For the default manuscript setting $r \leq 2$, this gives $\rho = 2$, hence $c_1 = 3$, $c_2 = 2$, i.e.,

$$K \leq 3R_0 + 2m(0).$$

9 Informational Mass–Energy Bookkeeping

To provide hardware designers with a register-level cost model, define an informational mass and a radiative energy register. In this section, the symbol m_0 is the discrete bookkeeping constant and is distinct from the dimensional calibration μ introduced in Section 5 (Equation (6)).

Definition 9.1 (Informational mass)**.** *Let $m_0 > 0$ be a model unit mass per unit positional weight. For a digit vector $V = (v_i)$ define the discrete informational mass*

$$M(V) := m_0 \sum_i |v_i|\, W_i. \tag{23}$$

Designers may couple the discrete mass $M(V)$ to continuous propagation resources via a substrate-dependent conversion constant λ and define an analog energy per arc $\mathcal{E}_n \propto \lambda L_n$ with $L_n = \frac{\pi}{2} F_n$. A continuous propagation budget can be defined as

$$\mathcal{B} := \kappa_B \sum_i |v_i| L_i = \kappa_B \frac{\pi}{2} \sum_i |v_i| F_i,$$

where κ_B is substrate-dependent. This budget links discrete informational mass and continuous trajectory resources and can be used to dimension photonic power, waveguide length, or analog gain.

10 Topological and Geometric Extension: Discrete–Continuous Transition and Wave–Particle Duality

10.1 Motivation and overview

The Specular Bit Architecture (SBA) as presented in previous sections is fundamentally discrete: information is encoded in signed ternary digits $\{-1, 0, +1\}$ with Fibonacci positional weights and exact local rewrite rules that guarantee deterministic normalization and bounded propagation. For selected substrates (notably photonic interconnects and neuromorphic waveguides), signal transport may be described with continuous propagation models [1]–[5]. This section defines an optional extension of SBA that embeds the discrete register geometry into a continuous topological propagation layer. The phrase "wave–particle duality" is used here in an engineering/modeling sense (coexisting discrete state updates and continuous transport parametrization), not as a replacement for standard quantum-mechanical formalism.

10.2 Assumptions specific to the extension

The results in this section rely on additional assumptions beyond the core discrete SBA model:

1) the geometric embedding is a design map from index space to physical coordinates, not a uniqueness theorem for physical layout;
2) continuum fields are coarse-grained effective variables obtained from local averages over many discrete updates;
3) coupling constants $\kappa_B, \lambda, G_I, \Lambda_I, \alpha, c_I$ are phenomenological and must be fitted per substrate;
4) Einstein-like closure in Equation (29) is a testable constitutive hypothesis rather than a consequence of the discrete rewrite algebra alone.

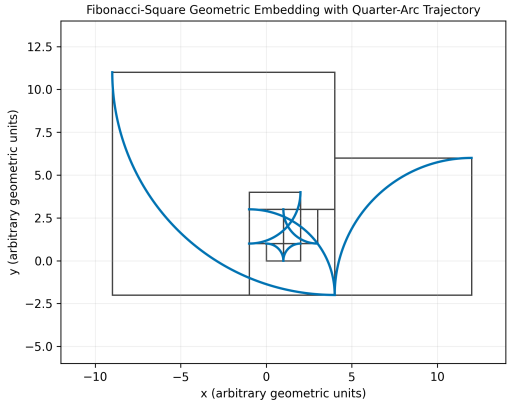

Fig. 1. Fibonacci-square geometric embedding with quarter-circle propagation arcs. The construction visualizes the discrete-to-continuous transition used in the topological extension.

10.3 Geometric embedding of Fibonacci registers

Let $(F_n)_{n \geq 0}$ denote the Fibonacci sequence with $F_0 = 0, F_1 = 1$. For each discrete register index n consider a square cell of side length F_n. We define the *storage area* associated to index n as

$$A_n := F_n^2,$$

which represents discrete information confinement at index n (memory cell, oligo block, or register tile).

Inside each square cell we inscribe a quarter-circle arc whose radius is proportional to the cell side. Choosing the radius $r_n = F_n$ (geometric scale matched to the discrete weight) the length of the quarter-circle arc is

$$L_n = \frac{1}{4} \cdot 2\pi r_n = \frac{\pi}{2} F_n.$$

This arc is used as the elementary continuous propagation vector associated with the n-th register: it parameterizes a local propagation segment connecting neighboring discrete cells.

10.4 Global wave trajectory and summation

The total continuous trajectory generated up to index n is the sum of the elementary arcs:

$$S_n = \sum_{i=1}^{n} L_i = \frac{\pi}{2} \sum_{i=1}^{n} F_i.$$

Using the identity $\sum_{i=1}^{n} F_i = F_{n+2} - 1$, one obtains the closed form

$$S_n = \frac{\pi}{2}\big(F_{n+2} - 1\big).$$

10.5 Asymptotic scaling and golden ratio

Because $\lim_{n\to\infty} F_{n+1}/F_n = \varphi$ (the golden ratio), the ratio of successive arc lengths satisfies

$$\lim_{n\to\infty} \frac{L_{n+1}}{L_n} = \lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi \approx 1.61803\ldots.$$

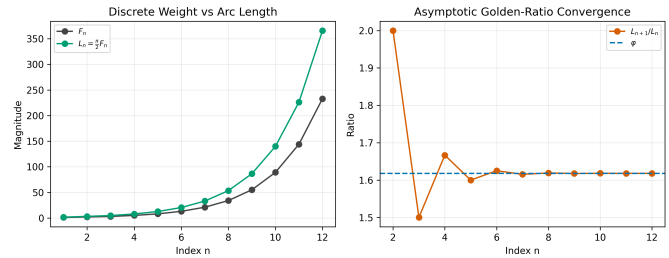

Fig. 2. Scaling consistency of the geometric model: left, Fibonacci weights and arc lengths $L_n = \frac{\pi}{2} F_n$; right, convergence of L_{n+1}/L_n to the golden ratio φ.

10.6 Implications for hardware design

1) **Layout principle.** Arrange memory tiles as Fibonacci-scaled squares; route waveguides or photonic interconnects along quarter-circle arcs inscribed in each tile as a geometry-informed layout option (to be validated with substrate-level electromagnetic analysis).
2) **Error resilience.** The discrete domain preserves exact data via canonical representations and local rewrites; in the geometric extension, propagation load is distributed along arcs by design.
3) **Adaptive redundancy.** Because arc lengths scale with F_n, one can allocate redundancy and analog gain proportionally to positional weight, implementing a Fibonacci-weighted protection scheme that concentrates resources where the lexicographic measure indicates higher informational mass.
4) **Neuromorphic mapping.** In neuromorphic photonic implementations, arc lengths and curvature provide a parameterization for delay-line and weighting structures; φ-scaling defines a multi-scale geometric hierarchy.
5) **Biological storage and the double-spiral hypothesis.** The quarter-circle propagation arcs $L_n = \frac{\pi}{2} F_n$ naturally induce spiral geometry in the embedding. If two independent specular SBA channels are implemented with opposite spatial orientation, the resulting geometry can be modeled as a dual intertwining spiral around a central axis. This provides a mathematically motivated design analogy for double-stranded DNA-like architectures in high-density molecular storage. In this manuscript, that link is treated as a hypothesis for substrate engineering and thermodynamic layout intuition, not as a claim that DNA helix formation is uniquely derived from SBA dynamics.

10.7 Mathematical remarks and consistency

1) The choice $r_n = F_n$ is a natural scale match; alternative proportionality constants $r_n = \kappa F_n$ ($\kappa > 0$) preserve the asymptotic φ-ratio while rescaling absolute arc lengths by κ.
2) The continuous summation S_n is compatible with the discrete informational mass $M(V) = m_0 \sum_i |v_i| W_i$ via a dimensional coupling factor: designers may calibrate a mapping constant λ such that analog

energy per arc $\mathcal{E}_n \propto \lambda L_n$ corresponds to modeled radiative energy increments in the bookkeeping model.

3) The embedding does not alter the formal correctness of the discrete rewrite system: all local rewrites and Protocol C arbitration remain defined on the discrete digit arrays. The geometric layer is an additional representational mapping that informs physical routing and energy provisioning.

10.8 Example: continuous propagation budget

Given a normalized discrete configuration with informational mass M, define a continuous propagation budget

$$\mathcal{B} := \kappa_B \sum_i |v_i| L_i = \kappa_B \frac{\pi}{2} \sum_i |v_i| F_i,$$

where κ_B is a substrate-dependent conversion constant. The budget $\mathcal{B}$ quantifies the total analog resource (e.g., photonic energy, waveguide length) required to transmit the stored information under the geometric embedding.

10.9 From informational mass to emergent spacetime curvature

We now give a formal bridge from discrete SBA occupancy to a continuum stress–energy source. The construction is model-level and substrate-agnostic: it specifies an effective geometric theory whose constants must be calibrated for any physical platform.

10.9.0.1 Step 1: coarse-grained information density from Fibonacci occupancy.: Let $\Omega \subset \mathbb{R}^3$ be partitioned into cells $\mathcal{C}_i$ associated with SBA indices i, and let

$$\omega_i := |v_i| W_i, \qquad \omega_i^{(u)} := K_i \, |v_i| W_i,$$

where $K_i \in \{0, 1\}$ flags unsanitized occupancy. Define piecewise-constant fields

$$\begin{aligned} \rho_I(x,t) &:= \frac{m_0}{\Delta V} \sum_i \omega_i(t)\, \chi_i(x), \\ \rho_u(x,t) &:= \frac{m_0}{\Delta V} \sum_i \omega_i^{(u)}(t)\, \chi_i(x), \end{aligned} \tag{24}$$

with χ_i the indicator of $\mathcal{C}_i$ and ΔV the cell volume.

10.9.0.2 Step 2: information current and continuity law.: Let u^μ be a coarse-grained timelike transport field (induced by local rewrite/reconstruction flow), and define information current

$$J^\mu := \rho_I u^\mu, \qquad \nabla_\mu J^\mu = -\Gamma_{\mathrm{san}}, \tag{25}$$

where $\Gamma_{\mathrm{san}} \geq 0$ is the local sanitization sink. In the post-commit effective limit (or on timescales where source/sink terms average out), $\nabla_\mu J^\mu \approx 0$.

10.9.0.3 Step 3: information potential generated by unsanitized gradients.: Define an effective scalar potential Φ_I by

$$\nabla^2 \Phi_I = 4\pi G_I \rho_u, \tag{26}$$

with G_I an effective information-gravity coupling. Within this model ansatz, unsanitized occupancy gradients act as a source term for weak-field curvature-like behavior.

10.9.0.4 Step 4: Information Complexity Tensor.: Let $g_{\mu\nu}$ be an emergent Lorentzian metric and c_I the effective propagation speed entering the continuum scaling. Define

$$T^{(I)}_{\mu\nu} = \left(\rho_I + \frac{p_I}{c_I^2} \right) u_\mu u_\nu + p_I g_{\mu\nu} + \Pi_{\mu\nu} + Q_{\mu\nu}, \tag{27}$$

where p_I is an effective information pressure,

$$Q_{\mu\nu} := \alpha \left(\nabla_\mu \rho_u \, \nabla_\nu \rho_u - \frac{1}{2} g_{\mu\nu} \nabla_\lambda \rho_u \nabla^\lambda \rho_u \right), \tag{28}$$

and $\Pi_{\mu\nu}$ is a traceless anisotropy tensor capturing directional rewrite-load asymmetry induced by Protocol C arbitration. The $Q_{\mu\nu}$ term is the explicit mathematical contribution of information-density gradients.

10.9.0.5 Step 5: Einstein-like closure for emergent metric dynamics.: Postulate the effective field equation

$$G_{\mu\nu} + \Lambda_I g_{\mu\nu} = \kappa_I T^{(I)}_{\mu\nu}, \qquad \kappa_I := \frac{8\pi G_I}{c_I^4}, \tag{29}$$

with Λ_I an optional background term. This closure should be interpreted as a constitutive hypothesis to be falsified/validated experimentally. By the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$, consistency requires

$$\nabla^\mu T^{(I)}_{\mu\nu} = 0, \tag{30}$$

which is the continuum conservation constraint corresponding to local SBA balance (rewrite exactness + deterministic commit + bounded sanitization source terms).

Proposition 10.1 (Continuum covariant conservation from SBA rounds)**.** *Assume: (i) synchronous SBA rounds with exact local rewrite preservation of weighted sum; (ii) Protocol C non-accumulation (at most one committed transfer per target per round, bounded magnitude in $\{-1, 0, 1\}$); (iii) finite second moments of local occupancy and transfer increments; (iv) existence of a hydrodynamic scaling limit $\Delta x, \Delta t \to 0$ with weak convergence of empirical fields (ρ_I, J^a, Π_{ab}); and (v) sanitization represented by a source density Γ_{san} that vanishes in the post-commit averaged limit (or is absorbed into a compensated current). Then the limiting Information Complexity Tensor of Equation* (27) *satisfies*

$$\nabla_\mu T^{(I)\mu}{}_\nu = 0.$$

Proof sketch. Under (i)–(ii), discrete cellwise balance for each round has conservative flux form

$$\Delta_t \rho_I^{(\Delta)} + \nabla_a^{(\Delta)} J^{a,(\Delta)} = -\Gamma_{\mathrm{san}}^{(\Delta)}.$$

Assumption (v) yields a source-free weak limit, giving $\nabla_\mu J^\mu = 0$. The first moment of the same balance law gives momentum/anisotropy transport with stress contributions from arbitration-induced directional loads and gradient regularization, converging (by (iii)–(iv)) to

$$\nabla_\mu T^{(I)\mu}{}_\nu = 0.$$

Equivalently, if Equation (29) holds, covariant conservation also follows from the contracted Bianchi identity. Hence SBA round dynamics imply continuum covariant conservation under assumptions (i)–(v). ∎

10.9.0.6 Weak-field/Newtonian limit.: For

$$g_{00} = -\left(1 + \frac{2\Psi}{c_I^2}\right), \qquad g_{ab} = \left(1 - \frac{2\Psi}{c_I^2}\right)\delta_{ab},$$

and slowly varying flow, Equation (29) reduces to

$$\nabla^2\Psi = 4\pi G_I \rho_{\rm eff}, \qquad \rho_{\rm eff} := \rho_I + \rho_\nabla + \rho_\Pi, \tag{31}$$

where ρ_∇ and ρ_Π are effective contributions from $Q_{\mu\nu}$ and $\Pi_{\mu\nu}$. In this effective description, unsanitized-bit accumulation contributes to curvature-like response through both occupancy and gradient-energy channels.

10.9.0.7 Interpretation.: Equations (24)–(31) provide the requested mathematical bridge:

1) discrete Fibonacci-weight occupancy $\sum_i |v_i| W_i$ induces coarse-grained density ρ_I;
2) unsanitized accumulation defines ρ_u and gradient stress terms;
3) these terms assemble into an Information Complexity Tensor $T^{(I)}_{\mu\nu}$;
4) under the closure hypothesis of Equation (29), metric response is modeled by Einstein-type dynamics $G_{\mu\nu} \propto T^{(I)}_{\mu\nu}$.

This yields a falsifiable program: estimate $(G_I, \Lambda_I, \alpha, c_I)$ from measured propagation delays, sanitization rates, and energy-register observables, then test whether inferred metric responses satisfy (29) across operating regimes with out-of-sample validation.

11 Testable Predictions and Experimental Protocols

This section converts the model into falsifiable numerical predictions using the illustrative calibration $\mu = 10^{-30}$ kg and round propagation at speed c. The predictions below are intentionally sharp: agreement supports the selected calibration/model closure; disagreement falsifies at least one modeling assumption. All thresholds are conditional on uncertainty budgets (instrumental, environmental, and model-form) stated by the experiment.

11.0.0.1 Decision-rule convention.: For each protocol, rejection should be based on a pre-registered criterion (confidence interval or Bayes factor) and accompanied by sensitivity analysis to nuisance parameters. Throughout this section, "falsified" means falsified within the specified uncertainty model, not disproven in an absolute sense.

11.1 Deep-space Casimir test beyond the Oort Cloud

11.1.0.1 Baseline (QED) reference.: For parallel plates (area A, gap a) at zero temperature,

$$P_{\rm QED}(a) = -\frac{\pi^2 \hbar c}{240 a^4}, \qquad F_{\rm QED} = A\, P_{\rm QED}. \tag{32}$$

With $a = 1\,\mu\mathrm{m}$ and $A = 1\,\mathrm{cm}^2 = 10^{-4}\,\mathrm{m}^2$:

$$P_{\rm QED} \approx -1.30 \times 10^{-3}\,\mathrm{Pa}, \qquad F_{\rm QED} \approx -1.30 \times 10^{-7}\,\mathrm{N}.$$

TABLE 1
Compact summary of quantitative predictions and falsification thresholds (illustrative calibration: $\mu = 10^{-30}$ kg, $r_\star = 3000$ AU, $a = 1\,\mu$m, $A = 1\,\mathrm{cm}^2$).

Observable	Baseline reference	SBA prediction	Falsification threshold
Casimir pressure at $r_\star$	$P_{\rm QED} \approx -1.30 \times 10^{-3}$ Pa, location-independent	$P_{\rm SBA} \approx -4.33 \times 10^{-7}$ Pa	Falsified if $P(a,r)$ stays location-independent within $< 1\%$ from 1 AU to $r_\star$
Casimir force at $r_\star$	$F_{\rm QED} \approx -1.30 \times 10^{-7}$ N, location-independent	$F_{\rm SBA} \approx -4.33 \times 10^{-11}$ N	Same $< 1\%$ location-independence criterion
Fractional Casimir drop (1 AU $\to r_\star$)	$\delta_{\rm drop} \approx 0$	$\delta_{\rm drop} = 1 - 1/3000 \approx 0.999666$ (99.9667%)	Any measured drop incompatible with $\Xi(r) = \mathrm{AU}/r$ at stated uncertainty
Single-event sanitization energy ($W_i = 34$)	No fixed $2\mu c^2 W_i$ event law	$\Delta E_{34} = 6.12 \times 10^{-12}$ J	Falsified by significant slope mismatch vs. Equation (35)
Thermal power at counted event rate	No fixed $P_{\rm th} \propto f$ map	$P_{\rm th} = f\,\Delta E_{34}$: 6.12×10^{-4} W at $10^8\,\mathrm{s}^{-1}$; 6.12×10^{-3} W at $10^9\,\mathrm{s}^{-1}$	Falsified if $E_{\rm meas}$ vs. N is non-linear (Equation (37))

11.1.0.2 SBA-topology correction.: Using the Fibonacci/geometric scaling of Section 10, define

$$n(r) := \frac{\ln(r/\mathrm{AU})}{\ln\varphi}, \qquad \Xi(r) := \varphi^{-n(r)} = \frac{\mathrm{AU}}{r}.$$

Model prediction for effective quantum-mode density and Casimir pressure:

$$\rho_{\rm q}^{\rm SBA}(r) = \rho_{\rm q}^{\oplus}\,\Xi(r), \qquad P_{\rm SBA}(a,r) = P_{\rm QED}(a)\,\Xi(r). \tag{33}$$

Choose a mission point beyond the Oort-cloud inner edge, e.g. $r_\star = 3000$ AU:

$$\Xi(r_\star) = \frac{1}{3000} = 3.333 \times 10^{-4}.$$

Hence:

$$P_{\rm SBA}(a, r_\star) \approx -4.33 \times 10^{-7}\,\mathrm{Pa},$$
$$F_{\rm SBA}(r_\star) \approx -4.33 \times 10^{-11}\,\mathrm{N}.$$

The predicted fractional drop relative to 1 AU is:

$$\delta_{\rm drop} = 1 - \Xi(r_\star) = 1 - \frac{1}{3000} = 0.999666\ldots\ (99.9667\%). \tag{34}$$

11.1.0.3 Falsification criterion.: After standard thermal/electrostatic/systematic corrections, if measured $F(a,r)$ is location-invariant within $< 1\%$ from 1 AU to $r_\star$, then Equation (33) with the above topology factor is falsified within that uncertainty budget. Conversely, non-invariance must also match the predicted $1/r$ trend within uncertainty to count as model support.

11.2 Thermal emission from mirrored-bit sanitization in an SBA photonic chip

11.2.0.1 Single-event energy prediction.: For mirrored occupancy cancellation at one index i, pre-sanitization occupancy is $+1$ and -1 at the same effective degree of freedom, so:

$$\Delta m_{\text{SBA}} = 2W_i, \qquad \Delta E_i = \mu c^2 \Delta m_{\text{SBA}} = 2\mu c^2 W_i$$

(from Equation (8)). With $\mu = 10^{-30}\,\text{kg}$, $c = 2.9979 \times 10^8\,\text{m}\,\text{s}^{-1}$:

$$\mu c^2 = 8.99 \times 10^{-14}\,\text{J}.$$

Thus:

$$\Delta E_i = 1.80 \times 10^{-13}\, W_i\ \text{J} = 1.12 \times 10^6\, W_i\ \text{eV}. \tag{35}$$

11.2.0.2 Prototype numeric target.: For a representative site with $W_i = 34$ (e.g., F_9):

$$\Delta E_{34} = 6.12 \times 10^{-12}\,\text{J} = 3.82 \times 10^7\,\text{eV}.$$

If sanitizations are clocked at event rate f, predicted thermalized power is:

$$P_{\text{th}} = f\,\Delta E_{34}. \tag{36}$$

Examples:

$$f = 10^8\,\text{s}^{-1} \Rightarrow P_{\text{th}} = 6.12 \times 10^{-4}\,\text{W},$$
$$f = 10^9\,\text{s}^{-1} \Rightarrow P_{\text{th}} = 6.12 \times 10^{-3}\,\text{W}.$$

For an effective thermal capacitance $C_{\text{th}} = 7 \times 10^{-3}\,\text{J}\,\text{K}^{-1}$:

$$\frac{dT}{dt} = \frac{P_{\text{th}}}{C_{\text{th}}} \approx 0.87\,\text{K}\,\text{s}^{-1} \quad (f = 10^9\,\text{s}^{-1}).$$

11.2.0.3 Falsification criterion.: Drive a counted mirrored-bit sanitization sequence N and measure emitted heat/radiation energy E_{meas}. The model predicts a linear law:

$$E_{\text{meas}} = N\,(2\mu c^2 W_i) + E_0, \tag{37}$$

with slope fixed by μ. A statistically significant slope inconsistent with Equation (35), or absence of the predicted linear scaling, falsifies the chosen (μ, sanitization-energy map) pair under the stated noise/systematics model.

11.3 Prime Signature Hypotheses under SBA Mirror Geometry

This subsection formalizes four hypothesis-level intuitions about prime numbers within the SBA mirror-geometry interpretation. These are not treated as proved number-theoretic statements; rather, they are posed as explicit, falsifiable signatures defined on finite encoded sequences.

Let $n \in \mathbb{N}$, and let $b(n) = (b_0, \ldots, b_{L-1}) \in \{0,1\}^L$ be a fixed-length binary encoding (left-padded to length L on each test batch). Let $x_k := 2b_k - 1 \in \{-1, +1\}$, and let the mirror sequence be $\bar{x}_k := -x_k$.

11.3.0.1 H1: absence of dominant sub-harmonic blocks (spectral simplicity).: Define the normalized periodogram

$$P_n(\omega_j) := \frac{1}{L}\left|\sum_{k=0}^{L-1} x_k e^{-2\pi i jk/L}\right|^2, \qquad \omega_j = \frac{2\pi j}{L}.$$

Define a sub-harmonic concentration score for $2 \le j \le \lfloor L/2 \rfloor$:

$$S_n := \frac{\max_{2 \le j \le \lfloor L/2 \rfloor} P_n(\omega_j)}{\sum_{j=1}^{\lfloor L/2 \rfloor} P_n(\omega_j)}.$$

Hypothesis H1: primes have lower median S_n than composites in matched bit-length strata.

11.3.0.2 H2: golden-spiral phase alignment.: Using SBA Fibonacci weights W_k, define weighted phase embedding

$$\theta_n := \operatorname{Arg}\left(\sum_{k=0}^{L-1} x_k W_k e^{ik\ln\varphi}\right),$$
$$r_n := \left|\sum_{k=0}^{L-1} x_k W_k e^{ik\ln\varphi}\right|.$$

Let $\mathcal{A} \subset [-\pi, \pi)$ be pre-registered angular sectors ("rays"). **Hypothesis H2:** prime phases θ_n are more concentrated in $\mathcal{A}$ than composite phases under the same L-stratified null.

11.3.0.3 H3: mirror chirality coherence (non-degenerate lag window).: Define mirror-overlap coherence for lag ℓ:

$$C_n(\ell) := \frac{1}{L-\ell}\sum_{k=0}^{L-\ell-1} x_k \bar{x}_{k+\ell}.$$

To avoid single-pair degeneracy at very large lags, fix a minimum overlap $m_{\min} \ge 4$ and restrict to

$$\mathcal{L}_{\text{eff}} := \{\ell \in \{1, \ldots, L-1\} :\ L - \ell \ge m_{\min}\}.$$

Define

$$\chi_n := \max_{\ell \in \mathcal{L}_{\text{eff}}} |C_n(\ell)|,$$

for samples with $L > m_{\min}$. **Hypothesis H3:** primes have higher χ_n than composites after controlling for parity and bit-length.

11.3.0.4 H4: terminal asymmetry and arc-jump statistic.: Define terminal asymmetry

$$A_n := |x_{L-1} - x_{L-2}|,$$

and geometric arc-jump proxy

$$J_n := \left|\theta_n - \theta_{n-1}\right|_{2\pi}$$

(circular distance). For odd $n > 2$, $x_{L-1} = +1$ by construction; thus parity alone is non-discriminative. **Hypothesis H4:** among odd numbers, primes exhibit a distinct distribution of J_n (or A_n, J_n jointly) compared to odd composites.

11.4 A provable wheel-unit invariant and a conservative empirical metric

To keep a strict separation between theorem-level and hypothesis-level claims, we introduce one arithmetic invariant that is provable and one empirical metric that is explicitly falsifiable.

11.4.0.1 Provable invariant (wheel units).: Let $M := 30 = 2 \cdot 3 \cdot 5$ and define

$$U_{30} := \{r \in \{0, \ldots, 29\} : \gcd(r, 30) = 1\}$$
$$= \{1, 7, 11, 13, 17, 19, 23, 29\}.$$

For every prime $p > 5$,

$$p \bmod 30 \in U_{30}.$$

Equivalently, any integer $n > 5$ with $n \bmod 30 \notin U_{30}$ is composite. This is a necessary condition (not sufficient), and it is strictly stronger than last-bit parity because it simultaneously excludes multiples of $2, 3, 5$.

11.4.0.2 Cautious empirical metric (within wheel units).: For fixed bit-length L, define the wheel-unit occupancy vector over U_{30}:

$$\pi_r^{(g,L)} := \Pr\big(n \bmod 30 = r \mid n \in g,\ n \bmod 30 \in U_{30}, \\ \mathrm{bitlen}(n) = L\big).$$

For $r \in U_{30}$ and $g \in \{\text{prime}, \text{odd composite}\}$, define the stratified divergence score

$$D_{\text{wheel}} := \frac{1}{|\mathcal{L}|} \sum_{L \in \mathcal{L}} D_{\text{JS}}\Big(\pi^{(\text{prime},L)} \,\|\, \pi^{(\text{comp},L)}\Big),$$

with D_{JS} the Jensen–Shannon divergence and $\mathcal{L}$ the pre-registered set of bit-length bins.

11.4.0.3 Exact falsification wording.: The empirical claim — "prime and odd-composite wheel-unit phase-space occupancies differ under the declared controls" — is rejected if either of the following holds under pre-registered analysis:

$$\widehat{D}_{\text{wheel}} \le \tau_{\text{wheel}} \tag{38}$$

(with τ_{wheel} fixed before observing outcomes), or the out-of-sample estimate fails replication across independent batches after multiplicity correction. Satisfaction of Equation (38) falsifies this metric-level signature only; it does not alter the formal SBA arithmetic results.

11.5 Fibonacci-prime sparsity: formal divisibility constraints and falsifiable SBA signatures

This subsection isolates theorem-level arithmetic facts from exploratory SBA-signature claims for Fibonacci-indexed candidates. Let F_n denote the Fibonacci sequence with $F_1 = F_2 = 1$.

11.5.0.1 Formal divisibility constraints (proved arithmetic layer).: A standard identity gives

$$\gcd(F_m, F_n) = F_{\gcd(m,n)}.$$

Hence, if $a \mid b$, then $F_a \mid F_b$. Therefore, if n is composite and $n \neq 4$, one can choose a proper divisor d with $2 < d < n$, yielding $1 < F_d < F_n$ and $F_d \mid F_n$; thus F_n is composite. The only composite-index exception is $n = 4$, where $F_4 = 3$ is prime. In particular,

$$F_n \text{ prime} \implies n \text{ prime, except } n = 4.$$

This implication is necessary but not sufficient: many prime indices still yield composite Fibonacci values (e.g., $F_{19} = 4181 = 37 \cdot 113$).

11.5.0.2 Falsifiable SBA-signature layer (empirical).: Define the candidate set $\mathcal{P}_N := \{n \le N : n \text{ prime}\}$ and label each index by $y_n := \mathbf{1}\{F_n \text{ is prime}\}$. Let ϕ_n be a pre-registered SBA phase/signature map computed from the binary representation of F_n (or from index-space encoding, fixed a priori). For a pre-registered test functional T, define the out-of-sample effect $\Delta_N := T(\{\phi_n : y_n = 1\}) - T(\{\phi_n : y_n = 0\})$, with controls for bit-length and magnitude bins.

TABLE 2
Prime-signature experimental protocol (hypothesis layer): metrics, controls, and rejection rules.

Hyp.	Test metric	Controls	Rejection rule
H1	S_n spectral concentration	Bit-length stratification; balanced classes	Reject if median difference Δ_1 not above pre-registered τ_1 with corrected significance
H2	Circular concentration of θ_n in sectors $\mathcal{A}$	Same L-bins; phase-randomized null; sector pre-registration	Reject if concentration gain Δ_2 is absent/opposite across holdout sets
H3	Mirror coherence χ_n	Parity and L-matched groups; lag-window fixed a priori	Reject if Δ_3 unstable or vanishes under bootstrap CI
H4	Arc-jump statistic J_n among odd n	Exclude $n = 2$; compare primes vs odd composites only	Reject if distributional separation (e.g., Wasserstein/KS effect) fails replication
H5	Fibonacci-indexed signature Δ_N (Section 11.5)	Prime-indexed set $\mathcal{P}_N$; compare F_n-prime vs F_n-composite in pre-registered bins	Reject if $\widehat{\Delta}_N \le \tau_{\text{fib}}$ (Equation (39)) or no out-of-sample replication

11.5.0.3 Exact falsification wording.: The empirical claim — "Fibonacci-prime indices carry a distinct SBA signature relative to prime-indexed Fibonacci composites under the declared controls" — is rejected if either

$$\widehat{\Delta}_N \le \tau_{\text{fib}} \tag{39}$$

with τ_{fib} fixed pre-analysis, or the out-of-sample effect fails replication across independent batches after multiplicity correction. Rejection under Equation (39) falsifies only this signature-level claim and does not modify the formal SBA arithmetic theorems.

11.5.0.4 Global falsification criteria.: For each H_k, define a pre-registered effect size Δ_k between prime and composite groups in matched L-bins, and uncertainty-adjusted threshold τ_k. The hypothesis is rejected when

$$\Delta_k \le \tau_k$$

(or has opposite sign) under the declared multiple-testing correction. A replication failure across independent batches also counts as rejection.

11.5.0.5 Interpretive status.: Any positive result in this subsection supports only the specified signature model; it does not prove a new primality theorem. Conversely, consistent null results falsify these particular mirror-geometry prime signatures while leaving the formal SBA arithmetic core unchanged.

11.5.0.6 Negative-result replication note.: A reproducible second-generation SBA-native prime/composite experiment (odd-only, bit-length matched for $L = 8$ to 20; feature definition in Appendix A.4) showed no discriminative signal in out-of-fold evaluation: $N = 163{,}988$ with $81{,}994$ primes and $81{,}994$ odd composites, global AUC $= 0.496953$, permutation-null mean AUC $= 0.499839$, permutation-null max AUC $= 0.505109$, and empirical $p = 0.916944$ (300 permutations, 5 folds), so under this protocol the tested mirror-geometry signatures are not supported and the null finding constrains this feature family only, without affecting the formal SBA arithmetic theorems.

12 Related Work

SBA builds on classical signed-digit arithmetic and Fibonacci positional systems [6]–[11]. For substrate context we reference DNA data storage and coding literature [12]–[18], biological proofreading mechanisms [19], [20], and neuromorphic hardware reviews and hardware exemplars [21]–[25]. The informational mass–energy bookkeeping is presented as a register-level modeling device and is not a claim about fundamental physics; it is informed by thermodynamic interpretations of information processing and computation [26]–[30]. The dissipative interpretation adopted in the appendices is aligned with non-equilibrium self-organization frameworks [31], while the substrate-agnostic computational framing is consistent with foundational simulation perspectives [32]. The wave-based geometric perspective used in Section 10 is supported by foundational matter-wave and interference literature [1]–[5].

Relative to prior art, the primary novelty claim here is the integrated package: specular signed carriers without an explicit sign bit, Fibonacci-identity exact local rewrites, deterministic per-target non-accumulating arbitration (Protocol C), and finite exhaustive local enumeration used to extract bounded-propagation constants. We therefore position SBA as a systems-level synthesis of known ingredients plus new coupling rules/proofs, rather than as a claim that any single component is unprecedented in isolation.

13 Discrete-to-Geometric Mapping: φ and π

The discrete Fibonacci index space admits natural geometric embeddings. Using Binet's formula and exponential radial sampling, the Fibonacci tiling can be associated with a logarithmic spiral whose growth factor is $\ln \varphi$. Geometric motifs (e.g., pentagonal arrangements) connect φ to π via $\varphi = 2\cos(\pi/5)$. These geometric observations motivate the choice of admissible offsets $o_A, o_B \geq 2$ to avoid resonant overlap in implementations that map indices to spatial coordinates.

14 Worked Examples

Two concise worked examples illustrate normalization and sanitization dynamics.

14.1 Example 1: Simple Normalization

Channel X:

$$A^{(X)} = (0,0,1,0,0), \qquad B^{(X)} = (0,0,1,0,0).$$

Intermediate:

$$d^{(X)} = (0,0,2,0,0).$$

Interior rewrite at index 2:

$$d_2 \leftarrow 0, \quad d_3 \leftarrow 1, \quad d_0 \leftarrow 1.$$

Result: $d^{(X)} = (1,0,0,1,0)$. Termination in one round.

TABLE 3
Round-by-round tracking for Example 1 (simple normalization).

Round/Phase	Local state evolution (channel X)
R1–Read	Snapshot: $d = (0,0,2,0,0)$.
R1–Compute	Overflow at index 2: schedule interior rewrite $d_2 \leftarrow d_2 - 2,\ d_3 \leftarrow d_3 + 1,\ d_0 \leftarrow d_0 + 1$.
R1–Write	Commit simultaneously: $d = (1,0,0,1,0)$.
R1–Sanitize	No 11 carrier event; flags remain zero.
R1–Scheduling & Arbitration	No flagged reconstructions; no candidates.
R1–Commit	No compensating injection; state unchanged.
R2–Read	$d = (1,0,0,1,0)$, with $\lvert d_i\rvert \leq 1$ for all i: stable, terminate.

14.2 Example 2: Sanitization and Conflict Resolution

Channel X:

$$A^{(X)} = (0,1,0,0,0), \qquad B^{(X)} = (0,1,0,0,0).$$

Intermediate:

$$d^{(X)} = (0,2,0,0,0).$$

Boundary rewrite at index 1 yields $d^{(X)} = (1,0,1,0,0)$. Suppose sanitization flags $K_0 = 1$ and $K_2 = 1$; candidate reconstructions produce $\tilde{v}_0 = +1, \tilde{v}_2 = 0$. Protocol C selects $v_0 = +1$, $v_2 = 0$. After applying compensating injection at next round, $d^{(X)} = (1,0,1,0,0)$ remains and normalization terminates.

TABLE 4
Round-by-round tracking for Example 2 (sanitization and arbitration).

Round/Phase	Local state evolution (channel X)
R1–Read	Snapshot: $d = (0,2,0,0,0)$.
R1–Compute	Boundary overflow at index 1: schedule boundary rewrite $d_1 \leftarrow d_1 - 2,\ d_2 \leftarrow d_2 + 1,\ d_0 \leftarrow d_0 + 1$.
R1–Write	Commit simultaneously: $d = (1,0,1,0,0)$.
R1–Sanitize	Assume transient carrier flags $K_0 = 1,\ K_2 = 1$.
R1–Scheduling & Arbitration	Candidates: $\tilde{v}_0 = +1,\ \tilde{v}_2 = 0$. Protocol C deterministically selects $v_0 = +1, v_2 = 0$.
R1–Commit	At most one commit per target; selected compensation is queued/committed for next-round semantics.
R2–Read/Write	After compensation handling, normalized state remains $d = (1,0,1,0,0)$, with $\lvert d_i\rvert \leq 1$: stable, terminate.

15 Exhaustive Canonical Patterns (Complete List)

For reproducibility, Table 5 lists the complete set of canonical length-five patterns used in the exhaustive local enumeration. Each pattern is a 5-tuple in $\{-1,0,+1\}$ with non-adjacency. The enumeration yields exactly 43 patterns.

15.0.0.1 Verification of the canonical pattern table.: The canonical width-5 pattern table was produced by enumerating all 5-tuples in $\{-1,0,1\}^5$ and filtering by the non-adjacency constraint $x_i x_{i+1} = 0$. As a verification step we (i) checked that the table contains exactly 43 distinct entries, (ii) confirmed that each listed 5-tuple satisfies the non-adjacency constraint, and (iii) validated that the union of the listed patterns equals the filtered enumeration set (no omissions or duplicates). These checks were performed deterministically and exhaustively on the finite width-5 search space; the aggregated counts reported in Section 15 follow directly from this verified table.

16 Exhaustive Pair Enumeration and Critical Pairs

Given the 43 canonical patterns per channel, ordered pairs of canonical patterns yield at most $43 \times 43 = 1849$ ordered pairs. The conceptual exhaustive check classifies ordered pairs according to whether they produce a center overflow $|d_{\text{center}}| = 2$ at the initial intermediate array.

For the default offset setting $o_A = o_B = 2$ and under canonical inputs, the enumeration yields exactly **162 critical ordered pairs** for which the center sum equals ± 2 and sanitization is invoked. Table 7 reports the aggregated distribution of critical pairs by patternA id (left operand).

16.0.0.1 Reproducibility statement: Methodologically, this section separates two evidence layers: (i) *formal proofs in text* (lemmas/theorems proved in Sections 3–8 and Appendix A) and (ii) *script-generated computational evidence* produced by the proofbench `scripts/sba_theory_proofbench.py`. The latter is archived as machine-readable artifacts in `outputs/` (notably `width5_transitions_full.csv` and `verification_report.txt`; see Appendix A.3). The aggregated counts reported here are empirical summaries extracted from those generated artifacts and are used to support, not replace, the formal proofs.

16.1 Numerical Extraction of $R_{\max}$ and K from Width-5 Enumeration

Let $\mathcal{T}$ denote the certified finite transition table produced by exhaustive width-5 enumeration under fixed offsets and Protocol C assumptions. Each row $\tau \in \mathcal{T}$ corresponds to one local seed configuration and records its synchronous-round evolution to a stable state (no $|d_i| = 2$).

16.1.0.1 Operational definitions on $\mathcal{T}$.: For each transition row τ:

- let $i_{\text{seed}}(\tau)$ be the seed index where the initial overflow/reconstruction event is centered;
- let $S_t(\tau)$ be the active index set after round t (indices touched by overflow/reconstruction effects);
- let $T(\tau)$ be the first round such that the local state is stable.

Define

$$R_{\max} := \max_{\tau \in \mathcal{T}} \max_{0 \le t \le T(\tau)} \max_{i \in S_t(\tau)} |i - i_{\text{seed}}(\tau)|,$$
$$K := \max_{\tau \in \mathcal{T}} T(\tau).$$

Thus $R_{\max}$ is the worst observed spatial spread and K is the worst observed round latency over the finite exhaustive set.

16.1.0.2 Deterministic computation procedure.: (1) Enumerate all canonical width-5 operand pairs (per channel) and construct $\mathcal{T}$. (2) Simulate synchronous rounds exactly as specified (Read/Compute/Write/Sanitize/Schedule/Commit) until stability for each row. (3) Record $T(\tau)$ and the maximum displacement from $i_{\text{seed}}(\tau)$. (4) Take global maxima to obtain $R_{\max}$ and K.

16.1.0.3 Reproducibility note.: Because $\mathcal{T}$ is finite, the procedure yields exact numerical values (not asymptotic estimates) once the certified transition table is materialized by the proofbench run. In this manuscript, these numerically extracted values are treated as computational evidence, while correctness claims about rewrite preservation, non-accumulation, and termination remain grounded in the formal proofs.

17 Notes on Experimental Prototyping

This section provides conceptual design considerations for implementers.

17.1 Substrate Primitives

- **Carrier implementation:** carriers may be realized as pairs of photonic modes, electronic states, or DNA base pairs. Implementations must ensure deterministic sanitization of the transient 11 state.
- **Local read/write rounds:** substrate must support synchronous read-then-write cycles or an equivalent atomic update primitive.
- **Atomicity and local locking (operational requirement):** the substrate must provide an atomic write primitive for committing the results of the Compute phase to the physical carriers and must support a local, per-target locking mechanism for arbitration. Concretely, the hardware or microcontroller layer must guarantee that (i) all writes in the Compute phase become visible simultaneously at the end of the Write phase (no partial visibility or intra-round cascading), and (ii) a target-level atomic lock can be acquired and released within a single round to prevent multiple concurrent commitments to the same target index. These primitives are required to implement the synchronous read-then-write semantics and the Protocol C arbitration assumptions used in the proofs.
  Implementation note: a per-target latch bank plus CAS-style lock bit (or a tiny local arbiter FSM) is sufficient. Resolve winner selection from a latched candidate set first, then perform a single lock-acquire-and-commit transaction so no select/lock race can occur.
- **Arbitration and energy registers:** Protocol C requires deterministic arbitration and a locally accessible energy register E_{rad} per neighborhood.
- **Operational enforcement of cross-channel independence:** hardware must enforce separate physical carriers for channels A and B or equivalent access control to prevent cross-writes.

TABLE 5
Canonical Patterns of Length Five (43 Patterns)

ID	Pattern
1	$(-1, 0, -1, 0, -1)$
2	$(-1, 0, -1, 0, 0)$
3	$(-1, 0, -1, 0, 1)$
4	$(-1, 0, 0, -1, 0)$
5	$(-1, 0, 0, 0, -1)$
6	$(-1, 0, 0, 0, 0)$
7	$(-1, 0, 0, 0, 1)$
8	$(-1, 0, 0, 1, 0)$
9	$(-1, 0, 1, 0, -1)$
10	$(-1, 0, 1, 0, 0)$
11	$(-1, 0, 1, 0, 1)$
12	$(0, -1, 0, -1, 0)$
13	$(0, -1, 0, 0, -1)$
14	$(0, -1, 0, 0, 0)$
15	$(0, -1, 0, 0, 1)$
16	$(0, -1, 0, 1, 0)$
17	$(0, 0, -1, 0, -1)$
18	$(0, 0, -1, 0, 0)$
19	$(0, 0, -1, 0, 1)$
20	$(0, 0, 0, -1, 0)$
21	$(0, 0, 0, 0, -1)$
22	$(0, 0, 0, 0, 0)$
23	$(0, 0, 0, 0, 1)$
24	$(0, 0, 0, 1, 0)$
25	$(0, 0, 1, 0, -1)$
26	$(0, 0, 1, 0, 0)$
27	$(0, 0, 1, 0, 1)$
28	$(0, 1, 0, -1, 0)$
29	$(0, 1, 0, 0, -1)$
30	$(0, 1, 0, 0, 0)$
31	$(0, 1, 0, 0, 1)$
32	$(0, 1, 0, 1, 0)$
33	$(1, 0, -1, 0, -1)$
34	$(1, 0, -1, 0, 0)$
35	$(1, 0, -1, 0, 1)$
36	$(1, 0, 0, -1, 0)$
37	$(1, 0, 0, 0, -1)$
38	$(1, 0, 0, 0, 0)$
39	$(1, 0, 0, 0, 1)$
40	$(1, 0, 0, 1, 0)$
41	$(1, 0, 1, 0, -1)$
42	$(1, 0, 1, 0, 0)$
43	$(1, 0, 1, 0, 1)$

TABLE 6
Default parameters used for manuscript-level exhaustive enumeration and scripts.

Parameter	Default value
Channel offsets in manuscript	$o_A, o_B \geq 2$ (default $o_A = o_B = 2$)
Script Fibonacci offset	`OFFSET_O = 2` (generation script default)
Weight law	$W_i = F_{i+o}$
Local window size	width-5 (indices $-2, -1, 0, 1, 2$)

TABLE 7
Distribution of critical pairs by patternA id (counts).

ID	Count	ID	Count
1, 2, 3	9 each	25, 26, 27	9 each
9, 10, 11	9 each	33, 34, 35	9 each
17, 18, 19	9 each	41, 42, 43	9 each
Total: 162 critical pairs			

17.2 Measurement and Validation

- **Functional verification:** exhaustive width-5 window tests to verify bounded propagation.
- **Robustness tests:** inject transient 11 patterns and verify deterministic reconstruction and non-accumulation under worst-case overlapping flags.
- **Calibration protocol (conceptual):** to map m_0 and c to substrate energy units, measure per-sanitization energy and per-reconstruction energy under controlled canonical inputs and fit model parameters accordingly.

18 Discussion, Limitations, and Future Work

- **Limitations:** the framework assumes synchronous rounds and deterministic local arbitration primitives. Asynchronous implementations require additional formal analysis. The mass–energy bookkeeping is a modeling device; mapping m_0 and c to physical energy units depends on the substrate and must be calibrated experimentally.
- **Future work:** formal machine-checked verification (Coq/Isabelle) of the bounded-propagation invariant, hardware prototypes implementing Protocol C, integration experiments with archival DNA layers, and thermodynamic calibration experiments inspired

by irreversible-computation bounds are natural next steps [26], [29], [30].

18.1 Reviewer-facing threats to validity

To clarify external and internal validity boundaries, we highlight four risk axes often raised in peer review:

- **Model-form risk:** continuum closure choices (Section 10) may fit some substrates and fail on others.
- **Parameter identifiability:** $(\mu, G_I, \Lambda_I, \alpha, c_I)$ may be partially confounded without multi-regime experiments.
- **Measurement risk:** Casimir and thermal protocols are sensitive to drift, parasitics, and environmental gradients.
- **Transferability risk:** proofs guarantee formal behavior of the discrete model; engineering implementations may violate assumptions (atomicity, timing, local lock semantics) unless explicitly enforced.

These risks do not invalidate the formal core; they delimit the evidentiary scope of physical interpretations and motivate staged validation.

18.2 Scalability Challenges

Scaling SBA from local windows to very large arrays introduces three coupled bottlenecks:

- **Synchronization pressure:** global round barriers become progressively expensive with increasing array diameter. Even when per-index work is $O(1)$, barrier latency and clock skew can dominate wall-clock throughput for large n.
- **Arbitration fan-in hotspots:** nonuniform input statistics can create spatially clustered flagged regions, increasing contention on target locks and local arbiters. This does not violate bounded propagation, but it can degrade effective throughput through repeated local conflicts.
- **Memory-bandwidth locality limits:** the model is local in logic, but practical implementations still require moving digit buffers, flags, and metadata. If physical placement does not preserve neighborhood affinity, interconnect traffic can dominate compute.

Practical scaling therefore requires hierarchical tiling: neighborhood-local schedulers, region-level asynchronous handshakes, and periodic global synchronization only for consistency checkpoints. This architecture preserves local correctness while reducing barrier frequency and contention amplification.

18.3 Thermal Noise Analysis (Model-Level)

For noisy substrates, let observed local digit state be $\hat{d}_i = d_i + \eta_i$, with η_i representing thermal/noise-induced perturbations in carrier readout or thresholding. A useful abstraction is a temperature-dependent local error probability p_T that a valid ternary carrier state is misclassified in one round. Under independent first-order approximation:

$$\mathbb{E}[N_{\text{false flag}}] \approx n\, p_T, \qquad \mathbb{E}[N_{\text{extra rounds}}] \propto p_T,$$

so false sanitization flags increase linearly with size for fixed p_T. Protocol C still enforces non-accumulation per target, but elevated p_T increases arbitration load and energy-register activity. Design implications are: (i) adaptive thresholds tied to online noise estimates, (ii) temporal majority filtering for flag generation, and (iii) local confidence tagging so low-confidence candidates are penalized in the deterministic score. These mitigations are consistent with irreversible-computation and information-thermodynamics constraints [26], [29], [30].

18.4 Plan for Quantum Validation

Given the manuscript's explicit distinction between informational formalism and physical claims, a staged validation path is appropriate:

1) **Classical baseline certification:** verify finite-state transition correctness and measured $(R_{\max}, K)$ on deterministic hardware emulation.
2) **Quantum-circuit embedding of local rules:** encode one SBA neighborhood update as reversible subcircuits (unitary blocks plus measurement for sanitization events), then compare transition statistics with classical ground truth [27], [33].
3) **Noise-channel sensitivity study:** run the embedded circuit under depolarizing/dephasing noise models and estimate an effective p_T-analog for quantum hardware, relating logical error rates to added normalization latency.
4) **NISQ-device pilot:** implement minimal instances on available noisy intermediate-scale quantum platforms, emphasizing reproducibility and uncertainty quantification rather than speedup claims [34], [35].
5) **Cross-platform falsifiability criteria:** predefine acceptance tests (state-transition fidelity thresholds, bounded-propagation preservation, and energy/bookkeeping consistency proxies) to separate model-valid behavior from hardware artifacts.

This plan keeps the framework scientifically conservative: quantum experiments are used to test representational and dynamical consistency, not to assert unproven physical equivalence.

18.4.0.1 Experimental disclaimer: Predictions that connect the informational bookkeeping to measurable physical effects are speculative and require dedicated experimental protocols. The present framework supplies a theoretical mapping; it does not constitute empirical proof of cosmological claims.

19 Conclusion

We presented a substrate-agnostic, mathematically rigorous framework for a Specular Bit Architecture intended as an informational layer for bio-photonic neuromorphic chips and DNA storage systems. The framework provides canonical encoding semantics, exact local rewrite rules based on Fibonacci positional weights, deterministic sanitization and arbitration policies, a lexicographic termination measure, and an informational mass–energy bookkeeping model. The manuscript explicitly separates formal informational claims from speculative physical interpretations and provides a clear roadmap for reproducibility, falsifiability, and formal verification.

Appendix A Formal proofs

This appendix collects formal proofs for the principal lemmas and theorems stated in the main text. For clarity, each argument is written in a structured style and references only objects defined in Sections 3–8.

A.1 Symbolic proof of the Fibonacci identity

Let $W_k = F_{k+o}$. Using $F_{n+1} = F_n + F_{n-1}$:

$$\begin{aligned} W_{i+1} + W_{i-2} &= F_{i+1+o} + F_{i-2+o} \\ &= (F_{i+o} + F_{i-1+o}) + F_{i-2+o} \\ &= F_{i+o} + (F_{i-1+o} + F_{i-2+o}) \\ &= 2F_{i+o} = 2W_i. \end{aligned}$$

A.2 Formal proof sketch of the Fibonacci identity (Coq/Lean style)

A.2.0.1 Statement.: Let $(F_n)_{n\geq 0}$ satisfy $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Define $W_i := F_{i+o}$ for a fixed offset $o \geq 0$. For all $i \geq 2$, prove

$$2W_i = W_{i+1} + W_{i-2}.$$

A.2.0.2 Lean/Coq-style context.: Work over $\mathbb{N}$ (or any semiring), with recurrence lemmas available:

$$F_{n+2} = F_{n+1} + F_n, \qquad F_{n+1} = F_n + F_{n-1}\ (n \geq 1).$$

A.2.0.3 Proof sketch (term/tactic structure).: Fix $i \geq 2$. Set $n := i + o$. It suffices to show $2F_n = F_{n+1} + F_{n-2}$ (index shift by definition of W).

1) From recurrence at n:

$$F_{n+1} = F_n + F_{n-1}.$$

2) From recurrence at n:

$$F_n = F_{n-1} + F_{n-2}.$$

3) Substitute this directly into Step 1 (no subtraction in $\mathbb{N}$):

$$F_{n+1} + F_{n-2} = F_n + F_{n-1} + F_{n-2}$$

equivalently, in assistant-friendly additive form,

$$F_{n+1}+F_{n-2} = F_n+F_{n-1}+F_{n-2} = F_n+F_n = 2F_n.$$

4) Rearranging (commutativity of $+$):

$$2F_n = F_{n+1} + F_{n-2}.$$

5) Undo the substitution $n = i + o$:

$$2F_{i+o} = F_{i+o+1}+F_{i+o-2} \iff 2W_i = W_{i+1}+W_{i-2}.$$

The side condition $i \geq 2$ guarantees all indices are valid.

A.2.0.4 Pseudo-script outline.: In assistant syntax, one can encode the same proof as:

1) define `W o i := Fib (i+o)` and state `2*W o i = W o (i+1) + W o (i-2)` with hypothesis $2 \leq i$;
2) unfold W and set $n := i + o$;
3) rewrite with Fibonacci recurrence at n and $n - 1$;
4) normalize arithmetic (ring/linear arithmetic tactics);
5) close by reflexive equality after index simplification.

A.3 Proof sketch of Zeckendorf uniqueness for positive support (Lemma)

Proof. Fix an offset $o \geq 2$ and define $W_i := F_{i+o}$. Consider nonnegative integers represented as

$$N = \sum_{i\geq 0} x_i W_i, \qquad x_i \in \{0, 1\}, \qquad x_i x_{i+1} = 0.$$

The goal is to prove both existence and uniqueness.

A.3.0.1 Existence.: Let $N > 0$. Apply the greedy procedure: choose j maximal with $W_j \leq N$, set $x_j = 1$, and replace $N \leftarrow N - W_j$. Repeat until the remainder is zero.

We must show that this procedure never requires adjacent selected indices. Suppose W_j is selected. Then the new remainder satisfies

$$0 \leq N - W_j < W_{j+1} - W_j = F_{j+o-1} = W_{j-1},$$

where the equality uses Fibonacci recurrence. Hence the next selected index is at most $j - 2$, proving non-adjacency. Because the remainder strictly decreases and is integer-valued, the algorithm terminates.

A.3.0.2 Uniqueness.: Assume there are two distinct admissible representations:

$$N = \sum_i x_i W_i = \sum_i y_i W_i,$$

with $x_i, y_i \in \{0, 1\}$, non-adjacent in each sequence. Let j be the largest index with $x_j \neq y_j$. Without loss of generality, $x_j = 1$, $y_j = 0$. Subtracting gives

$$W_j = \sum_{i<j} (y_i - x_i) W_i.$$

Taking absolute values,

$$W_j \leq \sum_{i<j} |y_i - x_i| W_i.$$

The right-hand side is maximized when all admissible lower indices are used; under non-adjacency this maximum is strictly smaller than W_j (standard Zeckendorf dominance bound; omitted derivation). Contradiction. Therefore no two distinct admissible representations exist.

Hence existence and uniqueness both hold. □

Appendix B: Exhaustive width-5 transition tables

A.4 Prime-signature experimental feature set

The second-generation SBA-native prime/composite experiment in Section 11.3 uses the following pre-specified features on odd integers with bit-length matching: Fibonacci-residue trajectory variance and span at moduli $5, 7, 11, 13$, Zeckendorf support-gap statistics (support count, mean/std/max gap), mirror-lag modular coherence at the same moduli, and control descriptors for bit balance and normalized run count.

The full $5^5 = 3125$-row width-5 transition table is not printed in this document to save space. It is generated deterministically by the proofbench script `scripts/sba_theory_proofbench.py`, which is the single reference generator used for manuscript evidence. For reproducibility, the run exports parameter metadata and artifacts in `outputs/`, including `width5_transitions_full.csv`, `verification_report.txt`, `symbolic_proof.txt`, `energy_examples.csv`, `toy_simulation.csv`, and `unit_test_report.txt`.

A.5 Proof sketch of signed canonical uniqueness via specular carriers (Lemma)

Proof. Let $c = (c_i)$ and $c' = (c'_i)$ be canonical carrier sequences with equal weighted value:

$$\sum_i \delta(c_i)W_i = \sum_i \delta(c'_i)W_i.$$

Define the difference digits

$$z_i := \delta(c_i) - \delta(c'_i) \in \{-2, -1, 0, 1, 2\}.$$

Then

$$\sum_i z_i W_i = 0.$$

Assume $z \not\equiv 0$. Let j be maximal with $z_j \neq 0$.

A.5.0.1 Step 1: sign normalization.: If necessary replace z by $-z$ so that $z_j > 0$. Then

$$z_j W_j = -\sum_{i<j} z_i W_i \leq \sum_{i<j} |z_i| W_i.$$

Because $z_j \in \{1, 2\}$, left side is at least W_j.

A.5.0.2 Step 2: canonical structure bound (sketch).: Each $\delta(c_i)$ and $\delta(c'_i)$ is canonical signed-digit valued; therefore high-index cancellation requires sufficient lower-index weight. But Fibonacci weights satisfy strict growth, and admissible lower-index combinations under canonical constraints cannot reach W_j once j is the maximal disagreement index.

Formally, the maximal lower-index canonical magnitude is bounded by a sum dominated by $W_j - 1$ (shifted Zeckendorf bound). Hence

$$\sum_{i<j} |z_i| W_i < W_j,$$

contradicting Step 1.

Thus $z \equiv 0$, i.e., $\delta(c_i) = \delta(c'_i)$ for all i, and therefore $c = c'$ pointwise.

For the specular swap operator σ, by definition it exchanges (p_i, n_i), so $\delta(\sigma(c_i)) = -\delta(c_i)$; applying twice restores the original pair, so σ is involutive. This preserves uniqueness under sign inversion. ☐

A.6 Preservation of weighted sum by interior rewrites (Equations (4)–(5))

Proof. Let $i \geq 2$, and consider the local contribution to

$$\Sigma(d) := \sum_k d_k W_k.$$

All untouched indices cancel in $\Delta\Sigma$, so only updated coordinates matter.

A.6.0.1 Case $d_i = +2$.: The rewrite is

$$d_i \leftarrow d_i - 2, \quad d_{i+1} \leftarrow d_{i+1} + 1, \quad d_{i-2} \leftarrow d_{i-2} + 1.$$

Hence

$$\Delta\Sigma = (-2)W_i + W_{i+1} + W_{i-2}.$$

Using $2W_i = W_{i+1} + W_{i-2}$, we get $\Delta\Sigma = 0$.

A.6.0.2 Case $d_i = -2$.: The rewrite is

$$d_i \leftarrow d_i + 2, \quad d_{i+1} \leftarrow d_{i+1} - 1, \quad d_{i-2} \leftarrow d_{i-2} - 1,$$

so

$$\Delta\Sigma = 2W_i - W_{i+1} - W_{i-2} = 0.$$

Therefore both interior rewrites are exactly weight-preserving. Since each rewrite is local and exact, any finite composition of such rewrites preserves $\Sigma(d)$. ☐

A.7 Protocol C non-accumulation (Lemma)

Proof. Fix one round and one target index j.

A.7.0.1 Deterministic candidate set.: By assumption, all proposals $v_{i\to j}$ are computed from the same latched snapshot. Therefore the proposal multiset for j is deterministic.

A.7.0.2 Deterministic winner selection.: Candidates are ordered by deterministic key $(E_{i\to j}, \text{tie-break})$. Hence there is a unique selected winner w_j.

A.7.0.3 Atomic commit serialization.: Only w_j is authorized to attempt commit for target j. Commit requires acquiring the target lock. Atomic lock semantics implies at most one linearized successful commit at τ_C for that target in that round.

A.7.0.4 Bounded value.: Every candidate is in $\{-1, 0, 1\}$; therefore the committed value (if any) is in $\{-1, 0, 1\}$.

Combining the above, the sum of committed transfers to fixed j in one round is exactly one element of $\{-1, 0, 1\}$, never a multi-source accumulation. This proves non-accumulation. ☐

A.8 Lexicographic descent and bounded propagation (Theorem)

Proof. Let

$$\begin{aligned} p(d) &:= (R(d), m(d)), \\ R(d) &:= \max\{i : |d_i| = 2\}, \\ m(d) &:= \big|\{i : |d_i| = 2\}\big|. \end{aligned}$$

Assume finite support and synchronous phases (Read, Compute, Write, Sanitization, Scheduling/Arbitration, Commit).

A.8.0.1 Step 1: right-edge control.: Compute rewrites act only at overflow sites and update indices $i-2, i, i+1$. Because writes are atomic at τ_W, there is no intra-round chaining beyond this precomputed set. Therefore no new overflow can appear strictly to the right of pre-round $R(d)$.

A.8.0.2 Step 2: multiplicity reduction at fixed right edge.: At index $R(d)$, acted rewrites remove the ± 2 overflow at that site. Sanitization/reconstruction may inject corrections, but Protocol C allows at most one commit per target and bounded magnitude, preventing same-round multi-accumulation that would recreate multiple right-edge overflows.

Hence after one round, either:

- rightmost overflow index decreases, or
- it stays equal and the number of overflows at/right of that edge decreases.

Equivalently, $p(d)$ decreases lexicographically.

A.8.0.3 Step 3: termination by well-foundedness.: $\mathbb{N} \times \mathbb{N}$ with lexicographic order is well-founded; thus infinite strictly descending chains do not exist. Therefore after finitely many rounds, there is no index with $|d_i| = 2$.

A.8.0.4 Step 4: bounded propagation.: Every operation has bounded spatial radius (rewrite stencil plus reconstruction radius r). Consequently influence cones are bounded by constants depending only on local rule radii and arbitration constraints, not on global register length n. Therefore there exist constants $R_{\max}, K$ independent of n such that normalization completes within radius $R_{\max}$ and at most K rounds for each finite local disturbance class.

This proves bounded propagation and bounded-round normalization. □

A.9 Informational mass monotonicity under local normalization

Proof. Recall

$$M(d) = m_0 \sum_k |d_k| W_k,$$

with $m_0 > 0$. For a single $+2$ rewrite at i, only $i, i+1, i-2$ change, so

$$\begin{aligned}\Delta M = m_0 \Big[&(|d_i - 2| - |d_i|) W_i \\ &+ (|d_{i+1} + 1| - |d_{i+1}|) W_{i+1} \\ &+ (|d_{i-2} + 1| - |d_{i-2}|) W_{i-2} \Big].\end{aligned}$$

The first bracketed term is nonpositive and at most $-2W_i$; each of the other two is nonnegative and at most $+W_{i+1}$, $+W_{i-2}$, respectively. Hence

$$\Delta M \le m_0 \big(-2W_i + W_{i+1} + W_{i-2} \big) = 0$$

by the Fibonacci identity.

Thus the worst-case local bound is non-increasing. The -2 rewrite is symmetric and yields the same estimate.

If a candidate-reconstruction policy in Protocol C includes a penalty for positive ΔM, then selected commits are biased toward non-increasing local mass, reinforcing dissipative normalization at the bookkeeping level. □

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